



- 4 A particle Q is initially positioned at a distance d vertically above a particle P . Particle P is projected with speed U at an angle α above the horizontal. At the same time, Q is projected at an angle β below the horizontal. Both particles move freely under gravity. The particles collide at time T after the projections.

(a) Show that $d = UT(\sin \alpha + \cos \alpha \tan \beta)$. [4]

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting or typing. There are no margins, text, or other markings on the page.



The particles collide when P is at its maximum height.

- (b)** Given that $\alpha = 30^\circ$ and $\beta = 60^\circ$, find d in terms of U and g . [3]

[illegible]